HW 2

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(a) Given $Z_1, \ldots, Z_n$ are i.i.d. Bernoulli(p) for $p \leq 1/2$ and $t \in (0, 1-p)$, we can compare the upper bound

$$P\left[\frac{1}{n}\sum_{i=1}^{n} Z_i \geq t + p\right]$$

between Hoeffding's inequality, Bernstein's inequality, and Bennett's inequality:

- **Hoeffding's inequality:** Since $Z_i \in [0,1]$, Hoeffding's inequality gives:
$$P\left(\frac{1}{n}\sum_{i=1}^{n} Z_i \geq t + p\right) \leq \exp(-2nt^2)$$
thus, $R_H = 2nt^2$.
- **Bernstein's inequality:** Define $X_i = Z_i - p$, then $E[X_i] = 0$ and $Var(X_i) = p(1-p)$. Since $|X_i| \leq 1 - p$, Bernstein's inequality gives:
$$\begin{aligned} P\left(\sum_{i=1}^{n} X_i \geq nt\right) &\leq \exp\left(-\frac{n^2t^2}{2np(1-p) + \frac{2}{3}(1-p)nt}\right) \\ &= \exp\left(-\frac{n^2t^2}{2n(1-p)(p+t/3)}\right) \\ &= \exp\left(-\frac{nt^2}{2(1-p)(p+t/3)}\right) \end{aligned}$$
thus, $R_{\text{Bern}} = \frac{nt^2}{2(1-p)(p+t/3)}$.

- **Bennett's inequality:** Using the same X_i as above, Bennett's inequality gives:

$$P\left(\sum_{i=1}^{n} X_i \geq nt\right) \leq \exp\left(-\frac{np(1-p)}{(1-p)^2} h\left(\frac{(1-p)nt}{np(1-p)}\right)\right)$$
$$= \exp\left(-\frac{np}{1-p} h\left(\frac{t}{p}\right)\right)$$

thus, $R_{\text{Benn}} = \frac{np}{1-p} h\left(\frac{t}{p}\right)$.

(b) We can compute R_{Bern}/R_H using (a):

$$\frac{R_{\text{Bern}}}{R_H} = \frac{\frac{nt^2}{2(1-p)(p+t/3)}}{2nt^2} = \frac{1}{4(1-p)(p+t/3)}$$

We can see if Bernstein's bound improves over Hoeffding's more than a constant factor if $R_{\text{Bern}}/R_H \to \infty$ as $n \to \infty$. Since $p \leq \frac{1}{2}$, $\frac{1}{2} \leq 1-p \leq 1$, so $1-p$ is bounded away from 0. For the ratio to approach infinity, we need $p + t/3 \to 0$ as $n \to \infty$. This only occurs when $p \to 0$ and $t \to 0$ as $n \to \infty$. Thus, Bernstein's bound improves over Hoeffding's more than a constant factor when $p \to 0$ and $t \to 0$ as $n \to \infty$.

(c) $R_{\text{Benn}}/R_{\text{Bern}}$ depends on (p, t) only through $x = t/p$:

$$\frac{R_{\text{Benn}}}{R_{\text{Bern}}} = \frac{\frac{np}{1-p} h\left(\frac{t}{p}\right)}{\frac{nt^2}{2(1-p)(p+t/3)}} = \frac{2p(p+t/3)}{t^2} h\left(\frac{t}{p}\right)$$
$$= \frac{2p(p+px/3)}{p^2x^2} h(x) = \frac{2(1+x/3)}{x^2} h(x)$$

Bennett's bound is never than Bernstein's since this ratio is always greater than 1:

$$\frac{2(1+x/3)}{x^2} h(x) \geq 1$$
$$h(x) \geq \frac{x^2}{2(1+x/3)}$$
$$\text{Let } f(x) = h(x) - \frac{x^2}{2(1+x/3)}$$
$$h(x) = (1+x)\log(1+x) - x \therefore$$
$$h'(x) = \log(1+x), h''(x) = \frac{1}{1+x} \therefore$$
$$f''(x) = \frac{1}{1+x} - \frac{1}{(1+x/3)^3} \geq 0$$

Since $f(0) = 0, f'(0) = 0, f''(x) \geq 0,\ f(X) \geq 0$ for all $x \geq 0$. Therefore, $h(x) \geq \frac{x^2}{2(1+x/3)}$, and substituting this into the ratio gives:

$$\frac{2(1+x/3)}{x^2}h(x) \geq \frac{2(1+x/3)}{x^2} \cdot \frac{x^2}{2(1+x/3)} = 1$$

thus, Bennett's bound is never worse than Bernstein's.
As $x \to 0$, the taylor expansion of $h(x)$ gives $h(x) = \frac{x^2}{2} - \frac{x^3}{6} + O(x^4)$, so

$$\frac{2(1+x/3)}{x^2}h(x) = \frac{2(1+x/3)}{x^2}\left(\frac{x^2}{2} - \frac{x^3}{6} + O(x^4)\right) = 1 + O(x)$$

Thus, as $x \to 0$, the ratio approaches 1, and Bennett's bound is asymptotically equivalent to Bernstein's.
As $x \to \infty$, the taylor expansion of $h(x)$ gives $h(x) = x\log(x) - x + O(\log(x)) \sim x\log(x)$, so

$$\frac{2(1+x/3)}{x^2}h(x) = \frac{2(1+x/3)}{x^2}(x\log(x)) \sim \frac{2}{3}\log(x) = \infty$$

Thus, as $x \to \infty$, the ratio approaches infinity, and Bennett's bound is asymptotically better than Bernstein's.

AI Usage: All homework problems were written in VSCode and compiled using LaTeX, with assistance from ChatGPT and Github Copilot inline suggestions.